Bagging, Boosting, and Bias–Variance on Olivetti Faces:

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Roll No. M26CSE018 · Machine Learning, Assignment 2

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Abstract

We study how decision-tree ensembles behave on a small, high-dimensional, 40-class face-recognition problem (Olivetti Faces; 240/80/80 train/val/test images, raw 4096-pixel inputs, no dimensionality reduction). We implement bootstrap aggregation (bagging) and multiclass SAMME boosting from scratch and evaluate them along four axes: accuracy, ensemble diagnostics (out-of-bag accuracy, pairwise disagreement, vote margin), robustness to a zeroed central 16×16 block, and empirical bias–variance over 30 stratified training resamples. The validation-selected single tree reaches 56.25% clean test accuracy; 100-tree bagging reaches 86.25%. Under the mask these fall to 7.50% and 67.50%, whereas the selected SAMME model (depth 4, 100 rounds) falls from 50.00% to 45.00%. Bagging cuts class-vector variance of the deep tree by 96% (0.650→0.026) while raising bias^2 (0.363→0.507); SAMME has the lowest variance (0.016) but the highest bias^2 (0.889) and the lowest resampled accuracy (28.2%). We explain several of these numbers analytically—e.g. a depth-d tree on balanced 40-class data peels off one identity per split, so its training accuracy is exactly $(d+1)/40$—and show that stability under resampling, stability under occlusion, and correctness are three distinct properties.

Keywords: decision trees, bagging, SAMME, face recognition, bias–variance, occlusion robustness, out-of-bag estimation.

1 Introduction

A decision tree is a low-bias, high-variance learner: given enough depth it can carve out one region per training image, but which regions it carves depends on exactly which images it saw. Two classical remedies attack this from opposite sides. *Bagging* [1] averages many trees grown on bootstrap replicates, which mainly reduces variance. *Boosting* [2] grows learners sequentially, re-weighting the training set so each new learner concentrates on previous mistakes; its multiclass form SAMME [3] is the one studied here. Face recognition with raw pixels is a deliberately hard testbed for both: there are 4096 features but only six training images per identity, so trees are forced to memorise, and an occluded region can invalidate precisely the pixels the trees rely on.

This report answers three questions with measurements rather than generic statements about the bias–variance trade-off:

Q1. How does a single tree change with depth, and what do bagging's diagnostics (OOB accuracy, disagreement, vote margin) say about the ensemble? When bagging is right or wrong, *what kind* of error is being fixed or kept?

Q2. How does a from-scratch SAMME ensemble behave—its weights, its hardest example, its cost, and its response to occlusion—relative to bagging?

Q3. What are the measured bias^2 and variance of the tree, bagging, and SAMME across 30 training resamples, how does bagging's benefit vary with depth, and is the lowest-variance model also the best?

Contributions. (i) Complete from-scratch bagging and SAMME implementations with explicit handling of the random-guessing threshold and perfect learners. (ii) A full set of ensemble diagnostics and a per-example error analysis. (iii) Closed-form explanations for several empirical patterns (tree leaf counts vs. training accuracy, the SAMME post-update weight mass, the OOB-vs-validation gap). (iv) A candid audit of what each measurement can and cannot support (Sec. 8).

2 Background and Notation

Bagging. Given n training pairs, tree $b \in \{1..B\}$ is fit on n indices drawn with replacement. A bootstrap draw contains $n[1-(1-1/n)^n] \approx 0.632n$ distinct images in expectation [4]; the remaining $\approx 0.368n$ are *out-of-bag* (OOB) for that tree. Prediction is a plurality vote $\hat{y}(x) = \arg\max_k \sum_b \mathbf{1}[h_b(x)=k]$. We use three diagnostics: *OOB accuracy* (each training image is classified only by trees that did not see it [5]); *pairwise disagreement* (mean over tree pairs of the fraction of images on which they differ); and *vote margin* (winning vote fraction minus runner-up fraction).

SAMME. For K classes, weights $w_i^{(1)} = 1/n$. At round t a depth-limited tree h_t is fit with sample weights $w^{(t)}$; with weighted error $\epsilon_t = \sum_i w_i^{(t)} \mathbf{1}[h_t(x_i) \neq y_i]$,

$$\alpha_t = \log\tfrac{1-\epsilon_t}{\epsilon_t} + \log(K-1), \quad w_i^{(t+1)} \propto w_i^{(t)} e^{\alpha_t \mathbf{1}[h_t(x_i)\neq y_i]}. \tag{1}$$

A learner with $\epsilon_t \geq 1 - 1/K$ is no better than random K-way guessing and is rejected; $\epsilon_t \to 0$ is guarded against an infinite α_t. Scores are $S_k(x) = \sum_t \alpha_t \mathbf{1}[h_t(x)=k]$ and the confidence is $q_k(x) = S_k(x)/\sum_t \alpha_t$.

Bias and variance of class vectors. Let $p_r(x_i) \in \mathbb{R}^K$ be the class vector of the model fitted on resample $r \in \{1..R\}$, $\bar{p}(x_i) = \frac{1}{R}\sum_r p_r(x_i)$ and e_i the one-hot label:

$$\text{Bias}^2 = \frac{1}{N}\sum_i \|\bar{p}_i - e_i\|^2, \quad \text{Var} = \frac{1}{NR}\sum_{i,r} \|p_{r,i} - \bar{p}_i\|^2. \tag{2}$$

By the usual expansion, $\text{Bias}^2 + \text{Var} = \frac{1}{NR}\sum_{i,r}\|p_{r,i} - e_i\|^2$ exactly (population variance). For a one-hot p

Table 1. Single-tree grid (accuracies in %). Depth 2–16 rows are identical for leaf sizes 1, 2, 4. "Occ." is mean training samples per leaf; "1-leaf" is the fraction of single-sample leaves.

Depth	Leaf	Train	Val	Test	Leaves	Occ.	1-leaf
2	1/2/4	7.50	7.50	5.00	3	80.0	0
4	1/2/4	12.50	10.00	7.50	5	48.0	0
8	1/2/4	22.50	20.00	12.50	9	26.7	0
16	1/2/4	42.50	36.25	25.00	17	14.1	0
None	1	100.00	60.00	57.50	48	5.0	16.7%
None	2	97.08	57.50	55.00	46	5.2	0
None	**4**	**96.67**	**63.75**	**56.25**	**40**	**6.0**	**0**

this equals $2(1 - \text{accuracy})$, which gives a free consistency check on our numbers (Sec. 6). Class vectors are `predict_proba` for a tree, vote fractions for bagging, and $q(x)$ for SAMME.

3 Experimental Setup

Data. The Olivetti Faces set [7] (via scikit-learn [8]) has 400 grayscale 64×64 images, 10 per identity, 40 identities. Inputs are the raw 4096-d pixel vectors (no PCA). The seed is 26018 (digits of the roll number). Within each identity, images are shuffled and split 6/2/2 into train/validation/test, giving 240/80/80 images; the same split is used everywhere. **Selection protocol.** Every hyper-parameter (tree depth and leaf size, bagging size B, boosting depth and rounds) is chosen on validation accuracy only and frozen; the test set is used for final reporting. **Occlusion.** A masked test set sets rows and columns 24–39 to zero (a 16×16 block, 6.25% of the pixels, covering the centre of the face). Models are trained on clean images only.

Models. *Tree:* scikit-learn `DecisionTreeClassifier` over depth $\in \{2, 4, 8, 16, \text{None}\} \times$ leaf $\in \{1, 2, 4\}$. *Bagging:* written by hand, $B \in \{1, 5, 10, 25, 50, 100, 200\}$, bootstrap size n, trees use the frozen Q1 setting. *SAMME:* written by hand with weighted trees of depth 1–4 and 10/25/50/100 rounds. *Q3:* 30 resamples, each drawing six images *with replacement within each identity* from that identity's six training images, test set fixed, no re-tuning. A resample contains on average $6[1 - (5/6)^6] \approx 3.99$ distinct images per identity (≈ 160 of 240), so every refit sees only about two thirds of the original information; this lowers all Q3 accuracies relative to the full-data models and should be kept in mind below.

4 Q1: Decision Tree and Bagging

4.1 Single tree: what changes with depth

Table 1 shows validation accuracy rising monotonically with depth, and the validation-best setting is the unrestricted tree with `min_samples_leaf`=4 (validation 63.75%, test 56.25%), which we freeze. Leaf size 1 fits the training set perfectly but validation (60.0%) is lower and 16.7% of its leaves hold a single image, i.e. it memorises individual photographs; a leaf size of 4 forbids this.

A structural regularity. For depth $d \leq 16$ the tree has exactly $d{+}1$ leaves and training accuracy exactly $(d{+}1)/40$ (3/40=7.5%, 5/40=12.5%, 9/40=22.5%, 17/40=42.5%). The exact match is consistent with each split isolating one whole identity (the Gini-favoured split on 40 balanced classes), i.e. each split "peels off" one person and the remaining leaf covers everyone else. A leaf predicts a single label, so a $(d{+}1)$-leaf tree can name at most $d{+}1$ identities and recognise at most $6(d{+}1)$ of the 240 training images—the bound is attained. Depth is therefore not a smooth complexity dial here: shallow trees are not "slightly underfit" but structurally unable to represent most classes. The unrestricted tree ends with 40 leaves, i.e. roughly one leaf per identity (6.0 images per leaf), which explains both its strong training fit and its brittleness: each identity is characterised by a handful of pixel thresholds.

4.2 Bagging implementation and bootstrap statistics

Each of the B trees is fit on $n{=}240$ indices drawn with replacement using the frozen setting (unrestricted depth, leaf 4). Trees are added incrementally so the first B trees of the 200-tree ensemble form every smaller ensemble. The expected number of distinct images per bootstrap is $240[1-(239/240)^{240}] = 151.9\ (\approx 0.633n)$. Its standard deviation is $\approx \sqrt{240(e^{-1} - 2e^{-2})} \approx 4.8$, so the average over $B{=}100$ draws should lie within about ±1 of 151.9 (notebook column `avg_distinct_training_examples`). Using validation accuracy, $B{=}100$ is selected: validation accuracy is 81.25/82.50/86.25% at $B{=}25/50/100$ (Table 7) and no tested size (including 200) exceeds 86.25%, since ties would have kept the smallest B.

Table 2 reports, for every B, the train, validation and test accuracy of the bagged ensemble together with the average number of distinct training images in one bootstrap sample and its theoretical expectation. Training accuracy reaches 100% from $B{=}25$ onward, while validation accuracy climbs from 32.50% ($B{=}1$) to 86.25% at $B{=}100$ and then drops to 83.75% at $B{=}200$; test accuracy keeps rising to 87.50% at $B{=}200$ but is not used for selection. The measured number of distinct images is 158.00 for the single draw at $B{=}1$ and approaches the expectation 151.89 as more draws are averaged (151.38 at $B{=}100$, 0.51 below it); every gap lies within about two standard errors ($\approx 4.8/\sqrt{B}$).

4.3 Do OOB accuracy, disagreement and margin tell the same story?

No. They measure different things (Fig. 1, Table 3). *OOB accuracy* rises from 0.24 ($B{=}1$) to 0.76 ($B{=}200$) and has not saturated. It is deliberately pessimistic relative to validation accuracy: at $B{=}1$ it is the accuracy of one tree on the $\approx$37% of images it never saw (0.24), far below the 63.75% validation accuracy of the full-data tree, because the bootstrap tree has seen only $\approx$3.8 distinct images per identity.

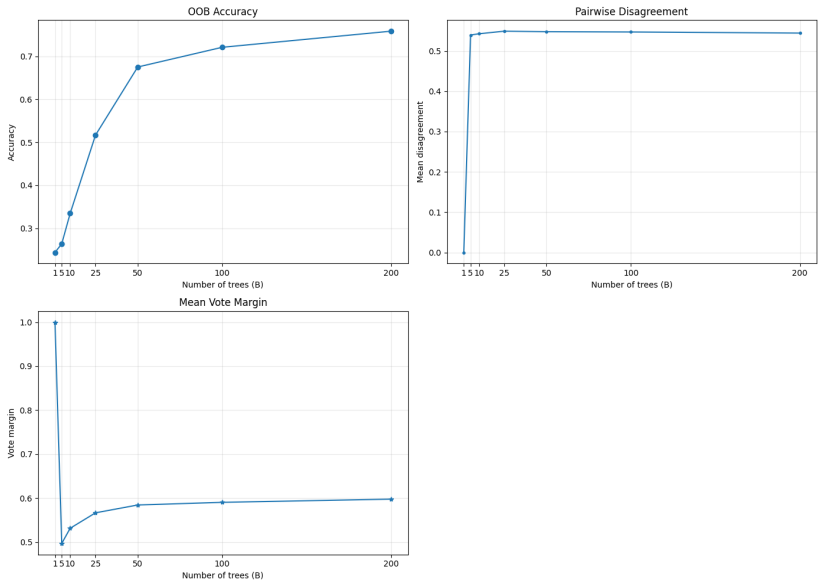


Figure 1. Q1. (a–c) OOB accuracy, pairwise disagreement and mean vote margin vs. number of trees B (the fourth panel of the export is intentionally blank). OOB accuracy keeps climbing to B=200 while disagreement is flat at ≈0.55 and the margin recovers slowly to ≈0.60.

Table 2. Bagging accuracy (%) and bootstrap statistics vs. B with the frozen tree setting. "Distinct" is the mean number of distinct training images in one bootstrap sample, averaged over the B draws; "Exp." is its expectation $240[1-(239/240)^{240}]$ and Δ = Distinct − Exp. The selected B is in bold.

B	Train	Val	Test	Distinct	Exp.	Δ
1	66.67	32.50	41.25	158.00	151.89	+6.11
5	93.33	58.75	47.50	155.40	151.89	+3.51
10	97.92	77.50	66.25	153.10	151.89	+1.21
25	100.00	81.25	80.00	152.36	151.89	+0.47
50	100.00	82.50	83.75	151.62	151.89	−0.27
100	**100.00**	**86.25**	**86.25**	**151.38**	**151.89**	**−0.51**
200	100.00	83.75	87.50	151.22	151.89	−0.67

At B=100 each image is voted on by only ≈37 trees, each trained on ≈63% of the data, so OOB (0.72) behaves like the accuracy of a ∼37-tree ensemble rather than a 100-tree one (validation at 25 and 50 trees is already 0.81–0.83). *Disagreement* jumps to ≈0.54 as soon as two or more trees exist and then stays flat: pairwise diversity is a property of the base learner and resampling scheme, not of B. Note it is measured on the training set, where about two thirds of the trees have seen the image (and tend to get it right) while the rest are wrong, so values near 0.55 mean that the out-of-bag votes are close to arbitrary for a single tree. High disagreement is therefore *necessary* for voting to help but not *sufficient*; what makes voting work is that wrong votes are scattered over many labels while correct votes concentrate on one. *Margin* is 1 at B=1 by construction, drops to 0.50 at B=5 (a few trees now compete) and slowly recovers to 0.60: with more trees, the plurality label's vote fraction becomes a more stable estimate. Hence accuracy improves, diversity is constant, and the margin is initially hurt before it improves—three different curves. Only validation accuracy is used to choose B.

Table 3. Bagging diagnostics on the training set vs. B (values read from Fig. 1, ±0.005). The last column is the expected number of trees for which a given training image is out-of-bag, $B(1-1/n)^n \approx 0.368B$.

B	OOB acc.	Disagree.	Margin	#OOB trees
1	0.243	0.000	1.000	0.4
5	0.264	0.540	0.497	1.8
10	0.335	0.543	0.532	3.7
25	0.517	0.549	0.567	9.2
50	0.675	0.548	0.585	18.4
100	0.721	0.547	0.591	36.8
200	0.759	0.545	0.598	73.6

Table 4. Q1(d)/Q2(d): clean vs. masked test accuracy (80 images; 95% Wilson interval in brackets) and mean winning probability / vote fraction / normalised weighted vote.

	Accuracy (%)		Mean confidence	
Model	Clean	Masked	Clean	Masked
Tree (leaf 4)	56.25 [45–67]	7.50 [4–15]	0.954	1.000
Bagging B=100	86.25 [77–92]	67.50 [57–77]	0.381	0.256
SAMME (d4, R100)	50.00 [39–61]	45.00 [35–56]	0.115	0.110

4.4 Clean vs. masked evaluation and confidence

The single tree loses 48.75 points (its correct count falls from 45 to 6 of 80 images) whereas bagging loses 18.75 (86.25→67.50%); bagging is +30.0 points better on clean and +60.0 points better on masked images. We read this

Table 5. Four clean test images (index, true id): single-tree prediction, bagging prediction, votes for the winner and for the true class (of 100).

Idx	True	Tree	Bag	Win votes	True votes	Outcome
2	1	31	1	58	58	bagging fixes
5	2	3	2	26	26	bagging fixes
9	4	39	2	20	8	both wrong
20	10	31	31	31	19	both wrong

as follows: the unrestricted tree's few early splits probably test central pixels, so zeroing that block misroutes nearly every image, whereas in a bagged ensemble only the subset of trees that use central pixels is misled. (We did not inspect the split features, so the first half of this explanation is a hypothesis.)

Confidence. The tree's mean winning class probability is 0.954 on clean data (accuracy 0.56) and *1.000 on masked data* (accuracy 0.075). The notebook computes `predict_proba` on the evaluated set; the value arises because every masked image falls into a pure leaf (leaves are nearly pure after fitting 96.7% of the training set), so the tree is maximally confident precisely when it is almost always wrong—its confidence carries no information about correctness. Bagging's winning vote fraction is not a calibrated probability (0.381 mean while 86% are correct, because votes spread over 40 labels), but it *does* move in the right direction under occlusion (0.381→0.256 as accuracy falls). SAMME's normalised vote barely changes (0.115→0.110). Only the bagging confidence is useful as a relative uncertainty signal.

4.5 Where bagging helps and where it cannot

Table 5 uses the first two images of each case type. **Variance-type error (fixed).** For image 2, the tree outputs identity 31, but only 4 of 100 bootstrap trees agree with it while 58 vote for the truth: the single tree's error was a consequence of the particular images it was fit on. For image 5 the true identity wins with only 26 votes (nearest rivals: 12 for identity 37, 11 for 27); a plurality among 40 classes needs a modest, not a large, margin. **Bias-type or ambiguity error (kept).** For image 20, identity 31 wins with 31 votes while the truth gets 19: a *consistent* pull toward the wrong identity across about a third of the trees, which no amount of averaging removes, because the evidence in the pixels favours the wrong person under these splits. For image 9 the vote is diffuse—the winner has only 20 votes, the truth 8, and 31 identities receive at least one vote—so the ensemble has not found any reliable evidence at all. Bagging removes errors that are idiosyncratic to one tree's training sample; it cannot create information the base learner never extracts.

Table 6. Clean validation accuracy (%) of SAMME by weak-learner depth and requested rounds. The selected cell is in bold.

Depth	10	25	50	100 rounds
1	5.00	5.00	13.75	12.50
2	6.25	11.25	16.25	23.75
3	16.25	27.50	31.25	32.50
4	20.00	28.75	41.25	**57.50**

5 Q2: Multiclass SAMME Boosting

5.1 One logged round on a toy problem

Six samples, labels (A, A, B, B, C, C), weights $1/6$, depth-1 tree predicting (A, A, B, B, B, B). The two C samples are wrong: $\epsilon = 1/3 < 1 - 1/3 = 2/3$, so the learner is accepted, $\alpha = \log 2 + \log 2 = 1.3863$, and after reweighting and normalisation the correct samples have weight $1/12$ each and the two errors $1/3$ each (sum 1). The same update is used on faces. **A useful identity:** since $e^\alpha = (K-1)(1-\epsilon)/\epsilon$, the normaliser is $\epsilon e^\alpha + 1 - \epsilon = K(1-\epsilon)$ and the previous errors hold exactly $(K-1)/K$ of the new weight, *independently of* ϵ (2/3 above; 0.975 for K=40). After every update the last learner would be exactly as good as random guessing on the new distribution; this is why $1 - 1/K$ is the rejection threshold, and why SAMME weights swing so violently (Sec. 5.3).

5.2 Validation grid and selection

Table 6 shows accuracy improving with depth and (mostly) with rounds; depth-1 stumps are the exception (12.5% at 100 rounds vs. 13.75% at 50). The best cell is depth 4 with 100 rounds (57.5%), selected on validation only, and all 100 learners were accepted. The optimum lies at the *corner* of the grid on both axes and the 50→100-round gain at depth 4 is large (41.25→57.5%), so the grid, as prescribed, probably under-explores deeper or longer boosting; we did not extend it so as to keep selection identical to the specification.

5.3 Round diagnostics

Fig. 2 supports a consistent picture. **Weighted error.** The first learner has $\epsilon_1 \approx 0.875 = 1 - 5/40$: an unweighted depth-4 tree has five leaves and recognises at most five identities (Sec. 4.1). Afterwards ϵ_t fluctuates around 0.6–0.7: each learner is genuinely weak, and by the identity above, *each fresh learner faces a distribution on which the previous one is random*. Weighted error is therefore not a measure of progress. **Learner weight.** $\alpha_t = \log \frac{1-\epsilon_t}{\epsilon_t} + \log 39$; the $\log 39 = 3.66$ offset keeps α_t positive until $\epsilon_t > 0.975$, so with $\epsilon \approx 0.65$, $\alpha \approx 3.04$, which matches the plateau at ≈3 (peak 3.87 at round 7 where $\epsilon = 0.45$). Because $\alpha_t \in [1.7, 3.9]$, SAMME's weighted vote here is close to an unweighted vote among 100 weak learners, each able to name only a handful of identities. **Weight concentration.** A misclassified sample's weight is multiplied by $e^\alpha \approx 20$

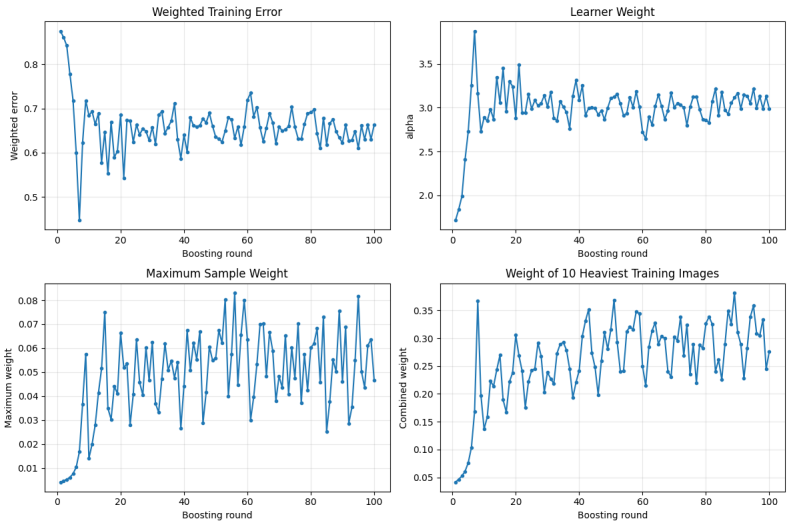


Figure 2. Diagnostics of the selected SAMME model over its 100 rounds: weighted training error, learner weight α_t, maximum sample weight and the combined weight of the 10 heaviest images.

before normalising; the maximum weight rises from the uniform 0.0042 to ≈0.08 and the 10 heaviest images carry up to ≈0.37 of total weight (uniform: 0.042), with violent round-to-round oscillation.

5.4 The hardest training face

Image 238 (identity 39) ends with weight 0.0688 and is misclassified by 88 of the 100 learners. Its 12 correct rounds (8, 13, 20, 21, 35, 42, 50, 57, 65, 75, 82, 89) recur every ≈7–8 rounds, a saw-tooth that follows from the update rule: while wrong, its weight grows ≈ 20× per round relative to correct samples, until it dominates the weighted Gini criterion and the next tree is forced to isolate its identity; once it is right it is no longer amplified, and its *normalised* weight collapses (the new errors now absorb 97.5% of the mass) until it has again been wrong for several rounds. Boosting thus keeps returning to this face because it never stays fitted. When wrong, learners most often predict identities 3 and 20 (10 times each), 14 (9), and 2, 4, 12 (8 each)—a short list that more plausibly reflects the label sets favoured by particular low-leaf trees than visual look-alikes (Fig. 3 shows an ordinary, well-lit smiling face, so there is no evidence of a label error or a defect). Persistent difficulty under a ≤16-leaf weak learner and 40 classes is the simplest explanation.

5.5 Bagging vs. SAMME: accuracy and cost

At every size bagging is far more accurate (test 80.0/83.8/86.3% vs. 17.5/31.3/50.0%), and the gap closes only slowly. Both reach 100% training accuracy at 100 learners, so perfect fit is not evidence of generalisation: boosting's train–test gap is 50 points at 100 rounds, bagging's is 14. The measured time is *higher* for bagging (70–136 s vs. 20–39 s), contrary to the usual folk wisdom that boosting is slower; but the comparison is confounded by the base learner (bagging trees are unrestricted and have ∼40 leaves, SAMME trees

Table 7. Bagging vs. SAMME (depth 4) at equal numbers of learners; times are single measured runs.

Method	L	Train	Val	Test	Time (s)
Bagging	25	100.00	81.25	80.00	70.0
SAMME	25	54.17	28.75	17.50	21.3
Bagging	50	100.00	82.50	83.75	136.1
SAMME	50	93.75	41.25	31.25	20.4
Bagging	100	100.00	86.25	86.25	125.7
SAMME	100	100.00	57.50	50.00	39.0

depth 4), the per-learner cost is not even stable across bagging runs (2.8, 2.7 and 1.3 s/tree), and bagging's independent fits are parallelisable while SAMME's are inherently sequential. We therefore claim only that, *in these implementations on this machine*, SAMME was cheaper per ensemble and much worse per learner.

5.6 Response to occlusion

SAMME loses the fewest points (50.00→45.00, −5.0) but starts from a lower base; bagging retains the highest absolute accuracy on both sets (Table 4). The 80-image test set means a 5-point drop is four images, On masked images the three Wilson intervals are disjoint (tree 4–15%, SAMME 35–56%, bagging 57–77%), so the ranking is resolved there; on clean images the tree and SAMME intervals overlap heavily. **A face with differing responses (test 5, Fig. 4).** On the clean image bagging predicts the truth (identity 2, 26 votes; runner-up 37 with 12) while SAMME is wrong (identity 3, score fraction 0.1098). Under masking, bagging stays correct but its lead shrinks to a single vote (identity 2: 14; identity 33: 13; identity 13: 12), so the correct answer has become fragile even though the label did not change; SAMME flips to the correct identity 2 with fraction 0.1056. The two models

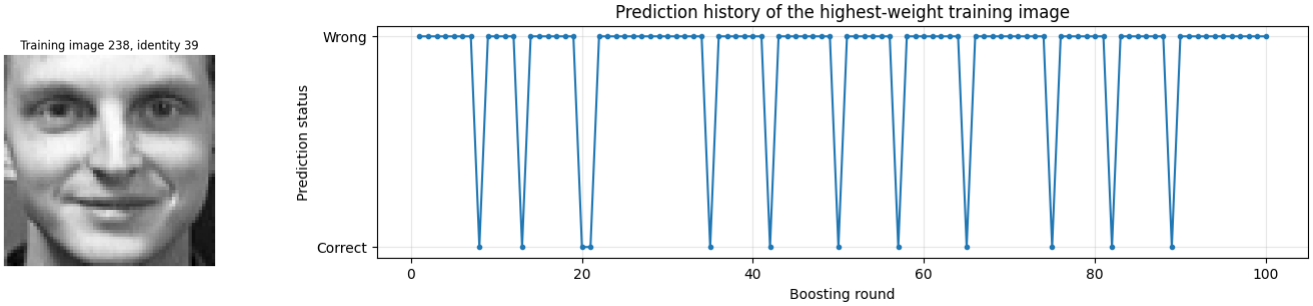


Figure 3. Training image 238 (identity 39), the largest final sample weight (0.0688, 16.5× uniform), and its per-round prediction history: wrong in 88 of 100 rounds.



Figure 4. Test image 5 (true identity 2), clean and masked.

Table 8. Bias^2, variance and test accuracy over 30 within-identity resamples (accuracy SD in percentage points).

Model	Bias^2	Var.	Acc. (%)	SD
Single tree	0.3629	0.6496	49.38	5.57
Bagging B=100	0.5071	0.0261	79.17	4.62
SAMME (d4, R100)	0.8885	0.0160	28.17	9.63

respond via different mechanisms. In bagging, the trees whose splits use the central block are scattered to random identities (the votes spread over more labels), but trees that do not use it are unaffected, so the plurality moves little; the margin is the casualty. In SAMME, learners have near-equal weights ($\alpha \approx 3$) and the winning fraction is only ≈ 0.11, so top scores are nearly tied and a few learners changing their output reverses the ranking—here helpfully, but this is as likely to happen in the other direction (SAMME's net accuracy still falls by 5 points). We did not inspect individual SAMME learners' votes on this image, so this account of the flip is an interpretation.

6 Q3: Empirical Bias and Variance

6.1 Selected models over 30 resamples

Sanity check. The tree's Bias^2 + Var $= 1.0125 = 2(1 - 0.4938)$, so its class vectors are effectively one-hot, as expected for near-pure leaves. For bagging the total is 0.533 and for SAMME 0.905. **Bagging** cuts variance by 96.0% (0.6496→0.0261) and accuracy rises by 29.8 points, but bias^2 *increases* by 0.144. Averaging does not reduce bias; the bagged trees are fit on bootstraps of resamples,

Table 9. Single trees and 100-tree bagging at three depths (leaf size 4). Acc. in %; Disag. is the mean pairwise disagreement of the 100 bagged trees; Leaves and Occ. (mean samples per leaf) are averages over the 30 refits (for bagging, over its trees).

Depth	Model	Bias^2	Var.	Acc.	Leaves	Occ.	Disag.
2	Single	0.9200	0.0370	5.58	3.0	80.0	—
	Bagged	0.9329	0.0112	12.54	3.2	76.7	0.961
8	Single	0.7994	0.1269	14.29	9.4	25.6	—
	Bagged	0.8122	0.0170	44.17	10.7	22.7	0.929
None	Single	0.3629	0.6496	49.38	40.0	6.0	—
	Bagged	0.5085	0.0258	77.21	39.5	6.1	0.825

i.e. on fewer distinct images, and are individually slightly worse. Net squared error still falls from 1.013 to 0.533. **SAMME** has the lowest variance (0.0160) and the largest bias^2 (0.8885) with the lowest accuracy (28.2%) and the largest accuracy SD (9.6 points). These are not contradictory. Variance measures how much the class *vector* moves, not whether it is right; SAMME's vector has maximum entry ≈0.11 and is nearly flat, so it can neither move much nor be close to a one-hot target: a true-class entry of just 0.05 already gives $\|p - e\|^2 \approx 0.9$, even when the argmax is correct. Hence the bias^2 and variance numbers of SAMME are not on the same scale as those of the tree (one-hot) or bagging (vote fractions) and cross-model comparisons of the two components must be read with that caveat; the accuracy column, which is scale-free, carries the actual ranking. SAMME is also the model most hurt by resampling (50%→28%): a plausible reason is that duplicated images add to the weight concentration already inherent in the algorithm (not tested here).

6.2 Depth: shallow, selected and deeper

Depth lowers bias and raises variance monotonically for single trees: bias^2 0.920→0.799→0.363 while variance rises 0.037→0.127→0.650. Leaf occupancy tracks this (80→26→6 images per leaf): the fewer images per leaf, the more a leaf's label depends on which images were drawn. Bagging removes 69.6%, 86.6% and 96.0% of the variance at depths 2, 8 and None, the largest share where the single tree

Table 10. Final summary: clean and masked test accuracy (%), accuracy drop in points, bias^2 and variance.

Model	Clean	Masked	Drop	Bias^2	Var.
Single tree	56.25	7.50	48.75	0.3629	0.6496
Bagging B=100	**86.25**	**67.50**	18.75	0.5071	0.0261
SAMME (d4, R100)	50.00	45.00	**5.00**	0.8885	**0.0160**

is most unstable. Bagging's bias^2 change is small at depths 2 and 8 (+0.013 each) but large at depth None (+0.146).

6.3 Is the deepest tree where bagging helps most?

No, but narrowly. The accuracy gains are +6.96 (depth 2), +29.88 (depth 8) and +27.83 points (unrestricted). The deepest tree has the largest *variance* reduction and by far the largest reduction in total squared error ($\Delta(\mathrm{Bias}^2+\mathrm{Var})$: 0.013, 0.097, 0.478), but not the largest *accuracy* gain. Reasons visible in Table 9: at depth 2 the 3-leaf trees can name only three identities, so their votes are individually near-random—disagreement is 0.961 but this is the disagreement of weak, uninformative voters, and bagging can lift accuracy only from 5.6% to 12.5% (and accuracy SD grows from 1.5 to 5.8 points). At depth 8 the trees are still diverse (0.929) but now carry enough information per leaf (23 images) for voting to recover many identities, and the single tree leaves 85.7% of the images wrong, i.e. headroom to gain. At unrestricted depth the single tree is already at 49.4%, so the achievable gain is capped, and disagreement is lowest (0.825) with 6 images per leaf: the trees agree more because they all fit nearly the same 40 identity-sized leaves. Two honest caveats: the difference between the depth-8 and unrestricted gains (2.0 points) is about 1.2 standard errors of its difference (accuracy SDs of 2.5–5.6 points over 30 resamples) and so is not decisive on its own, while the absolute accuracy ranking (77.2% for the deepest bagged model vs. 44.2%) is unambiguous.

7 Final Comparison and Discussion

Most stable. In the sense of variance across training resamples, SAMME (0.016), then bagging (0.026), then the tree (0.650); in terms of accuracy SD across resamples the order is bagging (4.6) $<$ tree (5.6) $<$ SAMME (9.6), so the two notions of "stable" disagree even for the same experiment. **Most sensitive to the missing region:** the single tree (48.75-point drop), followed by bagging (18.75) and SAMME (5.0). The tree's confidence stays at 1.0 as it fails. **Is the lowest-variance model the best?** No: SAMME has the lowest variance and the smallest drop yet the lowest resampled accuracy and the highest bias^2; bagging has the highest accuracy on both clean and masked images and on average over resamples, while its variance (0.026) exceeds SAMME's by only 0.010. The three notions—stability across training sets, stability under input perturbation, and correctness—must be reported separately. A low-variance model that is consistently wrong is stable and useless; SAMME's small mask drop partly reflects that it has little accuracy to lose.

8 Limitations and Threats to Validity

Sample size. Test accuracy is computed on 80 images (one image = 1.25 points); the 95% Wilson interval for 86.25% is 77–92%, and ranking of tree vs. SAMME on clean data (56.25% vs. 50.0%) is statistically unresolved. Results are from a single split and seed. **Resampling.** The 30 resamples draw from only six images per identity, so each fit has ≈4 distinct images per identity; the Q3 variance estimates describe sensitivity to this specific scheme, and resamples overlap heavily, so the 30 refits are not independent draws from the data distribution. **Vector scales.** Bias^2 and variance use different class-vector types (one-hot, vote fractions, normalised scores) and are not directly comparable across models (Sec. 6.1). Bagging in Q3(a) and Q3(b) use different random streams (bias^2 0.5071 vs. 0.5085 at identical settings), which is a rough indication of Monte-Carlo noise in these numbers. **Selection.** SAMME's optimum lies at the corner of its grid; bagging's B was chosen over a coarse grid. **Timing.** Training times are single runs on one machine with visible run-to-run variation. **Diagnostics.** Table 3 OOB, disagreement and margin values are read from the exported figure to ± 0.005 (the per-B CSV was not exported), and the empirical mean number of distinct bootstrap images is reported by the notebook rather than reproduced here. **Masking.** Only one synthetic occlusion at one position is tested; the mask is applied only at test time. Interpretive claims (why the tree collapses, why SAMME flips on test image 5) are hypotheses, flagged as such, that were not verified by inspecting split features or individual learners.

9 Conclusion

On this small face problem, bagging of unrestricted trees is the clear practical winner (86.25% clean, 67.50% masked, variance 0.026), SAMME with depth-4 learners is stable but weak (50.0%/45.0%), and the single tree is both unstable and fragile under occlusion (56.25%→7.50%, with confidence 1.0 while failing). Mechanistic observations explain much of this: a depth-d tree can name at most $d{+}1$ of the 40 identities; SAMME's update always leaves the last learner at random-guess level; and OOB accuracy lags validation accuracy because each image is scored by only $\approx 0.37B$ trees on ≈63% of the data. Finally, bagging's gain is not maximal at the deepest tree and its variance benefit does not imply a bias benefit: accuracy, stability under resampling, and stability under perturbation are separate properties that should be measured separately.

Reproducibility. Notebook, saved split and artifacts, result CSVs, figures, README and `requirements.txt` are in the repository at the commit above; seed 26018, split 6/2/2 per identity.

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